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United States Patent	12413131
Kind Code	B2
Date of Patent	September 09, 2025
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Method and apparatus for sensing inductor input/output current in a DC-DC converter circuit

Abstract

A switching regulator circuit has a switching transistor actuated during a switching on phase of a duty cycle. The current flowing through an inductor of the switching regulator circuit is determined from sensing a transistor current flowing through the switching transistor during switching on phase and dividing the sensed transistor current by the duty cycle to generate an output signal indicative of the inductor current of the switching regulator circuit. The duty cycle is determined from a detected ratio of switching transistor turn on time during the switching on phase and switching transistor turn off time during a switching off phase as controlled by the duty cycle control of the regulator circuit.

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Appl. No.: 18/225912

Filed: July 25, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230378864 A1	Nov. 23, 2023

Related U.S. Application Data

continuation-in-part parent-doc US 17748214 20220519 US 12003177 child-doc US 18225912

Publication Classification

Int. Cl.: H02M1/00 (20070101); H02M3/158 (20060101)

U.S. Cl.:

CPC H02M1/0009 (20210501); H02M3/158 (20130101);

Field of Classification Search

CPC: H02M (1/0009); H02M (3/158)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7202694	12/2006	Eberlein	N/A	N/A
7772818	12/2009	Smith	N/A	N/A
7795846	12/2009	Martin	N/A	N/A
7898825	12/2010	Mulligan	363/21.13	H02M 3/33507
8044644	12/2010	Huang et al.	N/A	N/A
8159203	12/2011	Derksen	323/284	H02M 3/158
8179105	12/2011	Lipcsei	N/A	N/A
8253405	12/2011	Trivedi et al.	N/A	N/A
8471545	12/2012	Couleur et al.	N/A	N/A
8482271	12/2012	Prodic	323/283	H02M 3/1588
8736244	12/2013	Voigtlander	N/A	N/A
9285399	12/2015	Babazadeh et al.	N/A	N/A
9292028	12/2015	Capodivacca et al.	N/A	N/A
9350158	12/2015	Chen et al.	N/A	N/A
9739810	12/2016	Childs	N/A	N/A
9755518	12/2016	Bansal et al.	N/A	N/A
9812963	12/2016	Nguyen et al.	N/A	N/A
10270342	12/2018	Pullen et al.	N/A	N/A
10454369	12/2018	Savic et al.	N/A	N/A
10756614	12/2019	Pullen et al.	N/A	N/A
10840803	12/2019	Toni et al.	N/A	N/A
10892683	12/2020	Nam	N/A	H02M 3/157
11193961	12/2020	Schrom et al.	N/A	N/A
11460873	12/2021	Nam et al.	N/A	N/A
2009/0322300	12/2008	Melanson	323/284	H02M 3/1563
2012/0019224	12/2011	Lima	323/282	H02M 3/1588
2013/0082675	12/2012	Capodivacca et al.	N/A	N/A
2015/0227146	12/2014	Knoedgen et al.	N/A	N/A
2018/0219484	12/2017	Mercer et al.	N/A	N/A
2018/0294804	12/2017	Ali	N/A	N/A
2018/0375435	12/2017	Muhoberac et al.	N/A	N/A
2019/0386561	12/2018	King et al.	N/A	N/A
2021/0067041	12/2020	Cho et al.	N/A	N/A

2021/0320590	12/2020	Ng et al.	N/A	N/A
2021/0351697	12/2020	Tyagi et al.	N/A	N/A
2022/0014102	12/2021	Sreenivas et al.	N/A	N/A
2022/0029520	12/2021	Chen et al.	N/A	N/A

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. patent application Ser. No. 17/748,214, filed May 19, 2022, the disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure generally concerns switching regulators such as DC-DC converters and, in particular, the measurement of inductor (input or output) current in a switching regulator.

BACKGROUND

(2) A switching regulator is a circuit that receives power at a first voltage level from a power source and outputs power at a second, different, voltage level for use by a downstream electronic system (the load). It is known in the art for a switching regulator to monitor input current in connection with carrying out various circuit operations and functions. For example, the switching regulator may include a current control loop that monitors the input current and uses the monitored input current as a control variable in a feedback loop to control the behavior of the switching regulator. It is also useful to be able to monitor the output current delivered to the load to determine, for example, the output power provided by the switching regulator.

(3) In a buck-type converter, for example, one known solution for determining the output current of the switching regulator is to sense the currents flowing through the high-side (HS) transistor switch and low-side (LS) transistor switch. Output current can then be calculated by summing the high-side and low-side currents. Those skilled in the art recognize, however, that the measurement of the low-side current is adversely affected by switching dead time at the switching node (see, FIG. 1 which shows buck converter operation where the dead times are on the LS transistor switch), during which no current through the HS or LS transistor switches can be sensed, despite the fact that the real current contribution is non-zero, and thus the summation of the high-side and low-side currents is an inaccurate measure of the actual output current of the switching regulator. The dead times in the switching node introduce a critical error in the summation-based high-side and low-side current measurement process. This error increases with increases in the switching frequency of the switching regulator.

(4) There is accordingly a need in the art for a method and apparatus that can more accurately determine the inductor (input or output) current of a switching regulator.

SUMMARY

(5) In an embodiment, a switching regulator circuit comprises: a high side (HS) transistor coupled between a first node and a switching node; a low side (LS) transistor coupled between the switching node and a second node; driver circuitry for controlling actuation of the HS transistor and LS transistor in accordance with a duty cycle including a switching on phase and a switching off phase delimited by dead time; and an inductor having a first terminal coupled to the switching node and a second terminal coupled to a third node, wherein a coil current flows through the inductor.

- (6) A coil current detection circuit of the switching regulator circuit comprises: a current sensing circuit configured to sense a transistor current flowing through only one of the HS transistor and LS transistor during the switching on phase and generate a first signal indicative of the sensed transistor current; and a duty cycle detect and divide circuit configured to detect said duty cycle which controls actuation of the HS transistor and LS transistor, and divide the first signal indicative of the sensed transistor current by said duty cycle to generate an output signal indicative of the coil current that flows through the inductor.
- (7) In an embodiment, a circuit for detecting a coil current flowing through an inductor of a switching regulator circuit having a switching transistor coupled to the inductor and driver circuitry for controlling actuation of the switching transistor in accordance with a duty cycle including a switching on phase and a switching off phase delimited by dead time comprises: a current sensing circuit configured to sense a transistor current flowing through the switching transistor during switching on phase of the duty cycle to generate a first signal indicative of the sensed transistor current; and a duty cycle detect and divide circuit configured to detect said duty cycle, and divide the first signal indicative of the sensed transistor current by said duty cycle to generate an output signal indicative of the coil current flowing through the inductor of the switching regulator circuit.
- (8) In an embodiment, a method for detecting a coil current flowing through an inductor of a switching regulator circuit having a switching transistor coupled to the inductor and driver circuitry for controlling actuation of the switching transistor in accordance with a duty cycle including a switching on phase and a switching off phase delimited by dead time comprises: sensing a transistor current flowing through the switching transistor during switching on phase to generate a first signal indicative of the sensed transistor current; detecting said duty cycle; and dividing the first signal indicative of the sensed transistor current by said duty cycle to generate an output signal indicative of the coil current flowing through the inductor of the switching regulator circuit.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The foregoing and other features and advantages will be discussed in detail in the following non-limiting description of specific embodiments in connection with the accompanying drawings, in which:
- (2) FIG. 1 shows a timing diagram for the voltage at the switching node of a buck-type switching regulator;
- (3) FIG. 2 is a block diagram showing a system including a buck-type switching regulator and an output current detection circuit that operates to determine the output current of the switching regulator;
- (4) FIG. 3 illustrates waveforms for currents and voltages at certain nodes of the switching regulator shown in FIG. 2;
- (5) FIG. 4 is a circuit diagram for a current sensing circuit of the output current detection circuit in FIG. 2;
- (6) FIGS. 5A and 5B are circuit diagram for embodiments of an amplifier used in a current sensing circuit;
- (7) FIGS. 6 and 7 are circuit diagrams for a duty cycle detect and divide circuit of the output current detection circuit in FIG. 2;
- (8) FIG. 8 is a block diagram showing a system including a boost-type switching regulator and an input current detection circuit that operates to determine the input current of the switching regulator;
- (9) FIG. 9 illustrates waveforms for currents and voltages at certain nodes of the switching regulator shown in FIG. 8;

(10) FIG. 10 is a circuit diagram for a current sensing circuit of the input current detection circuit in FIG. 8; and

(11) FIGS. 11 and 12 are circuit diagrams for a duty cycle detect and divide circuit of the input current detection circuit in FIG. 8.

DETAILED DESCRIPTION

(12) The same elements have been designated with the same reference numerals in the different drawings. In particular, the structural and/or functional elements common to the different embodiments may be designated with the same reference numerals and may have identical structural, dimensional, and material properties.

(13) Throughout the present disclosure, the term “connected” is used to designate a direct electrical connection between circuit elements with no intermediate elements other than conductors, whereas the term “coupled” is used to designate an electrical connection between circuit elements that may be direct, or may be via one or more intermediate elements.

(14) The terms “about”, “substantially”, and “approximately” are used herein to designate a tolerance of plus or minus 10%, preferably of plus or minus 5%, of the value in question.

(15) The determination of the output current of a buck-type switching regulator can advantageously be made dependent on a measurement of the current $I_{\text{sub.AVG}}(\text{HS})$ flowing through the high-side (HS) transistor switch during the switching on phase ($T_{\text{sub.ON}}$). The average current in the inductor of the switching regulator (i.e., the regulator output current) is then extrapolated solely from the sensed high-side transistor switch current using the switching duty cycle set by the ratio of the switching on phase ($T_{\text{sub.ON}}$) to the switching off phase ($T_{\text{sub.OFF}}$) for the HS transistor switch.

(16) Consider, for example, the following for a buck-type switching regulator. The average current $I_{\text{sub.AVG}}(\text{LS})$ flowing through the low-side (LS) transistor switch during the switching off phase ($T_{\text{sub.OFF}}$) is given by:

$$(17) I_{\text{AVG}}(\text{LS}) = I_{\text{AVG}}(\text{HS}) * \frac{T_{\text{OFF}}}{T_{\text{ON}}}$$

(18) The output current $I_{\text{sub.OUT}}$ of the switching regulator is then:

$$I_{\text{sub.OUT}} = I_{\text{sub.AVG}}(\text{HS}) + I_{\text{sub.AVG}}(\text{LS})$$

Substituting Gives:

$$(19) I_{\text{OUT}} = I_{\text{AVG}}(\text{HS}) * (1 + \frac{T_{\text{OFF}}}{T_{\text{ON}}}) = I_{\text{AVG}}(\text{HS}) * (\frac{T_{\text{ON}} + T_{\text{OFF}}}{T_{\text{ON}}})$$

(20) Where the real duty cycle D for switching the high-side (HS) transistor and low-side (LS) transistor switches is:

$$(21) D = \frac{T_{\text{ON}}}{T_{\text{ON}} + T_{\text{OFF}}}$$

(22) In this context, the “real” duty cycle means the duty cycle defined by the real times when the HS transistor switch is on ($T_{\text{sub.ON}}$) and when the LS transistor switch is on ($T_{\text{sub.OFF}}$), not the ideal duty cycle ($V_{\text{sub.OUT}}/V_{\text{sub.IN}}$).

(23) Simplified:

$$(24) I_{\text{OUT}} = \frac{I_{\text{AVG}}(\text{HS})}{D}$$

(25) Thus, by sensing the current $I_{\text{sub.AVG}}(\text{HS})$ flowing through the high-side (HS) transistor switch, and with knowledge of the real duty cycle D of the switching regulator, the output current of the switching regulator can be determined.

(26) The current $I_{\text{sub.AVG}}(\text{HS})$ can be measured through a power-power sense circuit (described in more detail below) and the real duty cycle D can be extrapolated from the ratio

$$(27) \frac{T_{\text{OFF}}}{T_{\text{ON}}}$$

through the drive signals which control switching of the high-side (HS) transistor and low-side (LS) transistor switches. A division of the measured current $I_{\text{sub.AVG}}(\text{HS})$ by the real duty cycle D of the switching regulator is then made by a detect and divide circuit (described in more detail below) to generate a value corresponding to the inductor output current of the switching regulator.

(28) Reference is now made to FIG. 2 which shows a block diagram for an output current detection

circuit **100** that operates to determine the output current of a switching regulator **102** (shown here to be of the buck-type, by example), and further to FIG. **3** which shows waveforms (in continuous conduction mode (CCM) operation) for currents and voltages at certain nodes of the buck-type switching regulator shown in FIG. **2**. The switching regulator **102** includes a high-side (HS) transistor switch **104** (for example, implemented as a p-channel MOS transistor) having a source terminal coupled, preferably directly connected, to an input voltage V_{in} and a drain terminal coupled, preferably directly connected, to a switching node **106**. The gate of the HS transistor switch **104** is driven by a high side driver circuit **108** that outputs a high side control signal **110** ($V_{gate.sub.HS}$). A current $I_{sub.HS}$ flows through the switch **104**. The switching regulator **102** further includes a low-side (LS) transistor switch **114** (for example, implemented as an n-channel MOS transistor) having a drain terminal coupled, preferably directly connected, to the switching node **106** and a source terminal coupled, preferably directly connected, to a reference voltage node (for example, ground). The gate of the LS transistor switch **114** is driven by a low side driver circuit **118** that outputs a low side control signal **120** ($V_{gate.sub.LS}$). A current $I_{sub.LS}$ flows through the switch **114**. In a preferred embodiment, the high side control signal **110** and the low side control signal **120** have a same frequency and are non-overlapping pulse width modulated (PWM) signals where the high side control signal **110** controls the HS transistor switch **104** to turn on during a switching on phase ($T_{sub.ON}$) and turn off during a switching off phase ($T_{sub.OFF}$). In an embodiment, the low side control signal **120** may be derived from the high side control signal **110** with a small margin (referred to as dead time **122**) to avoid both transistors being on at the same time. The low side control signal **120** controls the LS transistor switch **114** to turn on during a portion of the switching off phase ($T_{sub.OFF}$) which is delimited by the dead time **122** due to the non-overlapping switch control. An inductor **130** has a first terminal coupled, preferably directly connected, to the switching node **106** and a second terminal coupled, preferably directly connected, to an output node. A current $I_{sub.coil}$ flows through the inductor **130**. A load circuit (represented here by a load capacitance and resistance in parallel) is coupled to the output node.

(29) The output current detection circuit **100** includes a high side current sensing circuit **140** configured to sense the current $I_{sub.AVG(HS)}$, also referred to as the current $I_{sub.HS}$, flowing through the HS transistor switch **104** and output a signal indicative of the sensed current $I_{sub.AVG(HS).sub.sense}$. A duty cycle detect and divide circuit (1/D) **180** receives a high side control signal **110b** and a low side control signal **120b**. The signals **110b** and **120b** are derived from (or related to) the signals **110** and **120**, respectively, which control the switching on phase ($T_{sub.ON}$) and switching off phase ($T_{sub.OFF}$) but have a shorter dead time **122b** than the dead time **122** for the signals **110** and **120** (see, FIG. **2**) and are thus more accurately representative of the real duty cycle D of the switching regulator operation. The signal indicative of the sensed current $I_{sub.AVG(HS).sub.sense}$ is then divided by the determined real duty cycle D to generate an output signal that is indicative of the extrapolated value $I_{sub.OUTextrap}$ for the output current $I_{sub.OUT}$ of the switching regulator.

(30) Reference is now made to FIG. **4** which shows a circuit diagram for an embodiment of the current sensing circuit **140**. A first input transistor **M1** (for example, implemented as a p-channel MOS transistor) has a source terminal coupled, preferably directly connected, to the source terminal (s) of the HS transistor switch **104** (at the input voltage V_{in}) and a drain terminal coupled, preferably directly connected, to a first (inverting (-)) input of a differential amplifier **142**. The gate terminal of the first input transistor **M1** is driven by the high side control signal **110**. A second input transistor **M2** (for example, implemented as a p-channel MOS transistor) has a source terminal coupled, preferably directly connected, to the drain terminal (d) of the HS transistor switch **104** (at the switching node **106**) and a drain terminal coupled, preferably directly connected, to a second (non-inverting (+)) input of the differential amplifier **142**. The gate terminal of the second input transistor **M2** is also driven by the high side control signal **110**. Input transistors **M1** and **M2** are controlled by the high side control signal **110** to turn on during the switching on phase ($T_{sub.ON}$).

An output transistor M3 (for example, implemented as a p-channel MOS transistor) has a source terminal coupled, preferably directly connected, in a negative feedback circuit to the first (inverting) input of the differential amplifier 142. A resistor 144 with a fixed resistance R has a first terminal coupled, preferably directly connected, to a drain terminal of the output transistor M3 and a second terminal coupled, preferably directly connected, to the reference voltage node (for example, ground). A passive low-pass filter circuit 148 formed by the series circuit connection of a resistor 150 and capacitor 152 is coupled, preferably directly connected, between the drain terminal of the output transistor M3 and the reference voltage node. A series connection node 156 of the low-pass filter circuit 148 between the resistor 150 and capacitor 152 provides an output of the current sensing circuit 140 producing the signal V_{sense} indicative of the sensed current $I_{sub.AVG(HS).sub.sense}$. The first and second input transistors M1, M2 function to sample the source and drain voltages, respectively, of the HS transistor switch 104 responsive to the high side control signal 110 only during switching on phase (T.sub.ON) for application to the differential amplifier 142. The difference between those sampled voltages is converted by the differential amplifier 142 and output transistor M3 to a current $I_{sub.sense}$ which replicates the current flowing through the inductor 130 during switching on phase (T.sub.ON). This current $I_{sub.sense}$ is converted to a voltage by resistor 144 and the voltage is smoothed by the low pass filter circuit 148 to generate the output voltage signal V_{sense} .

(31) FIG. 5A shows a circuit diagram for an embodiment of the differential amplifier 142. A first amplifier transistor M4 (for example, implemented as a p-channel MOS transistor) has a source terminal coupled, preferably directly connected, to the inverting (−) input of the differential amplifier 142 and a drain terminal biased by a first current sink circuit 160 configured to sink a current $I_{sub.bias}$ to the reference voltage node. A second amplifier transistor M5 (for example, implemented as a p-channel MOS transistor) has a source terminal coupled, preferably directly connected, to the non-inverting (+) input of the differential amplifier 142 and a drain terminal biased by a second current sink circuit 162 configured to sink the current $I_{sub.bias}$ to the reference voltage node. The gate terminals of the transistors M4 and M5 are coupled, preferably directly connected, to each other and to the drain terminal of the transistor M4.

(32) Let K be a ratio of power to power sense. During the switching on phase (T.sub.ON) when the high side control signal 110 is asserted logic low, transistors 104, M1 and M2 are all turned on. If the gain of the differential amplifier 142 is high enough to ensure that the voltages at the inverting (−) input and non-inverting (+) input of the differential amplifier 142 are substantially equal (through the negative feedback with transistor M3), then the following is true:

$$V_{sub.106} - K * R_{sub.on} * I_{sub.bias} = V_{in} - K * R_{sub.on} * (I_{sub.bias} + I_{sub.sense})$$

(33) Where: $V_{sub.106}$ is the voltage at the switching node 106, $R_{sub.on}$ is drain-to-source resistance of the transistor 104 in the ON state (i.e., R_{ds_on}), $I_{sub.bias}$ is the bias current for the current sink circuits 160, 162, and $I_{sub.sense}$ is the current flowing through the transistor M3.

(34) Then, the sensed current $I_{sub.AVG(HS).sub.sense}$ is:

$$(35) I_{AVG(HS)}_{sense} = \frac{V_{in} - V_{106}}{K * R_{on}} = \frac{I_{out_on}}{K}$$

(36) Where: $I_{sub.out_on}$ is the average of the current flowing through the inductor 130 when the transistor 104 is in the ON state.

(37) It will accordingly be noted that the current $I_{sub.out_on}$ and its replica current $I_{sub.sense}$ (i.e., $I_{sub.AVG(HS).sub.sense}$) increase with a ratio of K. In this context, K is substantially constant, and it allows to have the current $I_{sub.sense}$ be directly proportional to the current flowing through the inductor 130 when the HS transistor switch 104 is in the ON state. It is important that the value for K is large enough to have the magnitude of $I_{sub.sense}$ in a range of some micro-amperes, despite an output current $I_{sub.out_on}$ that has a magnitude on the order of amperes.

(38) The signal at the output of the current sensing circuit 140 that is indicative of the sensed current $I_{sub.AVG(HS).sub.sense}$ is actually a low pass filtered voltage V_{sense} that is equal to

$R \cdot I_{\text{sub.AVG}}(\text{HS})_{\text{sub.sense}}$ where R is the resistance of the resistor **144**.

(39) Reference is now made to FIG. **6** which shows a circuit diagram for an embodiment of the duty cycle detect and divide circuit **180**. A differential amplifier **182** includes a non-inverting (+) input coupled to receive the voltage V_{sense} (which is indicative of the sensed current $I_{\text{sub.AVG}}(\text{HS})_{\text{sub.sense}}$ in the HS transistor switch **104** during the switching on phase ($T_{\text{sub.ON}}$)). The output of the differential amplifier **182** is coupled, preferably directly connected, to the inverting (−) input in negative feedback by a capacitor **184**. This forms a unity gain analog integration circuit that functions to integrate (i.e., average) the difference in voltages at the non-inverting (+) input and inverting (−) input. The voltage at the non-inverting (+) input is the voltage V_{sense} . The voltage at the inverting (−) input is a switched voltage selected by a voltage switching circuit **185** controlled by the ratio

$$(40) \frac{T_{\text{ON}}}{T_{\text{OFF}}}$$

to be equal to either the voltage $V_{\text{sub.current_out}}$ at the output of the differential amplifier **182** or the reference voltage (i.e., ground voltage).

(41) The voltage switching circuit **185** includes a first switching transistor M6 (for example, implemented as a p-channel MOS transistor) having a source terminal coupled, preferably directly connected, to the output of the differential amplifier **182** and a drain terminal coupled, preferably directly connected, to a switching node **186**. The first switching transistor M6 is controlled by signal **110b** to turn on during the $T_{\text{sub.ON}}$ state (i.e., when transistor **104** is turned on by the high side control signal **110**). Thus, the gate of transistor M6 may be configured to receive the control signal **110b** and is turned on when control signal **110b** is asserted (logic low). A second switching transistor M7 (for example, implemented as an n-channel MOS transistor) has a drain terminal coupled, preferably directly connected, to the switching node **186** and a source terminal coupled, preferably directly connected, to the reference voltage node. The second switching transistor M7 is controlled by signal **120b** to turn on during the $T_{\text{sub.OFF}}$ state (i.e., when transistor **104** is turned off by the high side control signal **110**). Thus, the gate of transistor M7 may be configured to receive the control signal **120b** and is turned on when control signal **120b** is asserted (logic high).

(42) The signal generated at the switching node **186** is equal, on average, to $D \cdot V_{\text{sub.current_out}}$, where $V_{\text{sub.current_out}}$ is the voltage at the output of the differential amplifier **182**. When the first switching transistor M6 is turned on during the switching on phase ($T_{\text{sub.ON}}$), the voltage $V_{\text{sub.current_out}}$ at the output of the differential amplifier **182** is applied to the inverting (−) input of the differential amplifier **182**. The differential amplifier **182** will respond to this by driving the voltage $V_{\text{sub.current_out}}$ toward the voltage $V_{\text{sense}} - V_{\text{sub.current_out}}$. Conversely, when the second switching transistor M7 is turned on during the switching off phase ($T_{\text{sub.OFF}}$), the reference voltage (ground) is instead applied to the inverting (−) input of the differential amplifier **182**. The differential amplifier **182** will respond to this by driving the voltage $V_{\text{sub.current_out}}$ toward the voltage V_{sense} . Over time, due to the averaging function provided by the unity gain integration on capacitor **184**, the voltage $V_{\text{sub.current_out}}$ at the output of the differential amplifier **182** will settle at a value equal to:

$$(43) V_{\text{current_out}} = V_{\text{sense}} * \left(\frac{T_{\text{ON}} + T_{\text{OFF}}}{T_{\text{ON}}} \right) = \frac{V_{\text{sense}}}{D}$$

(44) In other words, supposing that the voltage at the (+) and (−) inputs of the amplifier **182** are equal, the following equation applies:

$$(45) V_{\text{current_out}} = V_{\text{sense}} * \frac{1}{D} = \frac{R}{K} \frac{1}{D} I_{\text{HS}} = \frac{R}{K} I_{\text{OUT}}$$

(46) Where: R is the resistance of resistor **144** of FIG. **4**.

(47) The voltage $V_{\text{sub.current_out}}$ at the output of the differential amplifier **182** is the signal indicative of the extrapolated value $I_{\text{sub.OUT}}\text{extrap}$ for the output current $I_{\text{sub.OUT}}$ of the switching regulator.

(48) Reference is now made to FIG. **7** which shows a circuit diagram for another embodiment of the duty cycle detect and divide circuit **180**. A differential amplifier **192** includes a non-inverting

(+) input coupled to receive the voltage Vsense (which is indicative of the sensed current I.sub.AVG(HS).sub.sense in the HS transistor switch **104** during the switching on phase (T.sub.ON)). The output of the differential amplifier **192** is coupled, preferably directly connected, to the inverting (−) input in negative feedback by a resistor **194** having a fixed resistance Rb. The inverting (−) input is further coupled to ground through a resistor **196** having a variable resistance Ra. The value of the variable resistance Ra is digitally controlled by the bits of a digital selection signal **198** generated by an analog-to-digital converter (ADC) circuit **200**. This circuit is essentially a non-inverting operational amplifier with a controllable gain dependent on the resistances Ra and Rb (where the amplifier gain

$$(49) 0G = (1 + \frac{R_b}{R_a})).$$

(50) A voltage switching circuit **202** includes a first switching transistor M8 (for example, implemented as a p-channel MOS transistor) having a source terminal coupled, preferably directly connected, to a supply voltage Vdd and a drain terminal coupled, preferably directly connected, to a switching node **204**. The first switching transistor M8 is controlled by signal **110b** to turn on during the T.sub.ON state (i.e., when transistor **104** is turned on by the high side control signal **110**). Thus, the gate of transistor M8 may be configured to receive the control signal **110b** and is turned on when control signal **110b** is asserted (logic low). A second switching transistor M9 (for example, implemented as an n-channel MOS transistor) has a drain terminal coupled, preferably directly connected, to the switching node **204** and a source terminal coupled, preferably directly connected, to the reference voltage node (ground). The second switching transistor M9 is controlled by signal **120b** to turn on during the T.sub.OFF state (i.e., when transistor **104** is turned off by the high side control signal **110**). Thus, the gate of transistor M9 may be configured to receive the control signal **120b** and is turned on when control signal **120b** is asserted (logic high). The switching node **204** is coupled to the input of the ADC circuit **200** through a low-pass filter circuit formed by resistor **206** and capacitor **208**.

(51) The voltage at the switching node **204** is a switched voltage selected by the voltage switching circuit **202** controlled by the ratio

$$(52) \frac{T_{ON}}{T_{OFF}}$$

to be equal to either the supply voltage Vdd or the reference voltage (i.e., ground voltage). Analog filtering of the switched voltage by the low-pass filter circuit produces an input voltage to the ADC circuit **200** equal, on average, to D*Vdd. This switched and filtered voltage is converted by the ADC circuit **200** to generate the digital selection signal **198** for controlling the variable resistance Ra of the resistor **196**. The resistances Ra, Rb of the resistors **196**, **194** set the gain of the differential amplifier **192** applied to the voltage Vsense (indicative of the sensed current I.sub.AVG(HS).sub.sense in the HS transistor switch **104** during the switching on phase (T.sub.ON)) which is applied to the non-inverting (+) input.

(53) The following relationship is achieved through the non-inverting amplifier circuit using the differential amplifier **192** with resistors **194** and **196**:

$$(54) V_{\text{current_out}} = V_{\text{sense}}(1 + \frac{R_b}{R_a})$$

(55) The variable resistance Ra of the resistor **196** is digitally controlled to satisfy the following relationship:

$$(56) 1 + \frac{R_b}{R_a} = \frac{1}{D}$$

(57) Because Ra must satisfy the foregoing equation, this means that:

$$(58) R_a = \frac{D}{1-D} R_b$$

(59) Where the resistance Rb is fixed. If we consider, as an example, the ADC circuit **200** with a three-bit resolution, the input signal D*Vdd is then quantized to eight levels, with the following Table defining the relationship of the variable resistance Ra as a function of the duty cycle and the fixed resistance Rb:

(60) TABLE-US-00001 D Ra 0 0 1/8 Rb/7 2/8 Rb/3 3/8 3*Rb/5 4/8 Rb 5/8 5*Rb/3 6/8 3*Rb 7/8 7*Rb

(61) The output of the ADC circuit **200** controls the variable resistor **196** to select the correct resistance R_a . This can be accomplished, for example, using an analog multiplexer having mux inputs connected to resistances for the possible values of R_a , a control input connected to the output of the ADC circuit **200** and a mux output connected to resistor R_b and the input of amplifier **192**.

(62) It will be noted that the implementation with a three-bit resolution is just an example, and this concept is extendible to a higher resolution to provide higher accuracy in the selection of the variable resistance value.

(63) Because of the foregoing relationship between D , R_a and R_b , the following equation applies:

$$(64) V_{\text{current_out}} = V_{\text{sense}} * \frac{1}{D} = \frac{R}{K D} I_{HS} = \frac{R}{K} I_{OUT}$$

(65) Where: R is the resistance of resistor **144** in FIG. **4**.

(66) The voltage $V_{\text{sub.current_out}}$ at the output of the differential amplifier **192** is the signal indicative of the extrapolated value $I_{\text{sub.OUT}}^{\text{extrap}}$ for the output current $I_{\text{sub.OUT}}$ of the switching regulator.

(67) FIG. **3** illustrates operation of the buck converter in continuous conduction mode (CCM). It will, however, be noted that the current sensing circuit **140** and duty cycle detect and divide circuit **180** can also operate to sense inductor output current during ON time when the converter is instead configured for operation in discontinuous conduction mode (DCM). In this case, beyond the ON and OFF times, a high-impedance state is present when the coil current is zero. In this high-impedance state: transistors **104**, **114** are turned off, transistors **M6**, **M7** are turned off, and transistors **M8**, **M9** are turned off.

(68) The foregoing technique for buck-type regulator circuit output current determination is also applicable in the context of determining the input current of a boost-type regulator circuit.

(69) The determination of the input current of a boost-type switching regulator can advantageously be made dependent on a measurement of the current $I_{\text{sub.AVG}}(LS)$ flowing through the low-side (LS) transistor switch during the switching on phase ($T_{\text{sub.ON}}$). The average current in the inductor of the switching regulator (i.e., the regulator input current) is then extrapolated solely from the sensed low-side current using the switching duty cycle set by the ratio of the switching on phase ($T_{\text{sub.ON}}$) to the switching off phase ($T_{\text{sub.OFF}}$) for the LS transistor switch.

(70) Consider, for example, the following for a boost-type switching regulator. The average current $I_{\text{sub.AVG}}(HS)$ flowing through the high-side (HS) transistor switch during the switching off phase ($T_{\text{sub.OFF}}$) is given by:

$$(71) I_{AVG}(HS) = I_{AVG}(LS) * \frac{T_{OFF}}{T_{ON}}$$

(72) The input current $I_{\text{sub.IN}}$ of the switching regulator is then:

$$I_{\text{sub.IN}} = I_{\text{sub.AVG}}(HS) + I_{\text{sub.AVG}}(LS)$$

Substituting Gives:

$$(73) I_{IN} = I_{AVG}(LS) * (1 + \frac{T_{OFF}}{T_{ON}}) = I_{AVG}(LS) * (\frac{T_{ON} + T_{OFF}}{T_{ON}})$$

(74) Where the real duty cycle D for switching the high-side (HS) transistor and low-side (LS) transistor switches is:

$$(75) D = \frac{T_{ON}}{T_{ON} + T_{OFF}}$$

(76) In this context, the “real” duty cycle means the duty cycle defined by the real times when the LS transistor switch is on ($T_{\text{sub.ON}}$) and when the HS transistor switch is on ($T_{\text{sub.OFF}}$), not the ideal duty cycle ($V_{\text{sub.OUT}}/V_{\text{sub.IN}}$).

(77) Simplified:

$$(78) I_{IN} = \frac{I_{AVG}(LS)}{D}$$

(79) Thus, by sensing the current $I_{\text{sub.AVG}}(LS)$ flowing through the low-side (LS) transistor switch, and with knowledge of the real duty cycle D of the switching regulator, the input current of the switching regulator can be determined.

(80) The current $I_{\text{sub.AVG}}(LS)$ can be measured through a power-power sense circuit (described

in more detail below) and the real duty cycle D can be extrapolated from the ratio

$$(81) 0 \frac{T_{ON}}{T_{OFF}}$$

through the drive signals which control switching of the low-side (LS) transistor and high-side (HS) transistor switches. A division of the measured current $I_{sub.AVG}(LS)$ by the real duty cycle D of the switching regulator is then made by a detect and divide circuit (described in more detail below) to generate a value corresponding to the input current of the switching regulator.

(82) Reference is now made to FIG. 8 which shows a block diagram for an input current detection circuit 300 that operates to determine the input current of a switching regulator 302 (shown here to be of the boost-type, by example), and further to FIG. 9 which shows waveforms (in continuous conduction mode (CCM) operation) for currents and voltages at certain nodes of the switching regulator shown in FIG. 8. The switching regulator 302 includes a high-side (HS) transistor switch 304 (for example, implemented as a p-channel MOS transistor) having a source terminal coupled, preferably directly connected, to an output node and a drain terminal coupled, preferably directly connected, to a switching node 306. The gate of the HS transistor switch 304 is driven by a high side driver circuit 308 that outputs a high side control signal 310 ($V_{gate.sub.HS}$). A current $I_{sub.HS}$ (also referred to as the regulator output current $I_{sub.OUT}$) flows through the switch 304 to an output node. A load circuit (represented here by a load capacitance and resistance in parallel) is coupled to the output node. The switching regulator 302 further includes a low-side (LS) transistor switch 314 (for example, implemented as an n-channel MOS transistor) having a drain terminal coupled, preferably directly connected, to the switching node 306 and a source terminal coupled, preferably directly connected, to a reference voltage node (for example, ground). The gate of the LS transistor switch 314 is driven by a low side driver circuit 318 that outputs a low side control signal 320 ($V_{gate.sub.LS}$). A current $I_{sub.LS}$ flows through the switch 314. In a preferred embodiment, the high side control signal 310 and the low side control signal 320 have a same frequency and are non-overlapping pulse width modulated (PWM) signals where the low side control signal 320 controls the LS transistor switch 314 to turn on during a switching on phase ($T_{sub.ON}$) and turn off during a switching off phase ($T_{sub.OFF}$). In an embodiment, the high side control signal 310 may be derived from the low side control signal 320 with a small margin (referred to as dead time) to avoid both transistors being on at the same time. The high side control signal 310 controls the HS transistor switch 304 to turn on during a portion of the switching off phase ($T_{sub.OFF}$) which is delimited by a dead time 322 due to the non-overlapping switch control. An inductor 330 has a first terminal coupled, preferably directly connected, to the switching node 306 and a second terminal coupled, preferably directly connected, to receive an input voltage V_{in} . A current $I_{sub.coil}$ (also referred to herein as an input current $I_{sub.IN}$) flows through the inductor 330.

(83) The input current detection circuit 300 includes a low side current sensing circuit 340 configured to sense the current $I_{sub.AVG}(LS)$, also referred to as the current $I_{sub.LS}$, flowing through the LS transistor switch 314 and output a signal indicative of the sensed current $I_{sub.AVG}(LS)_{sub.sense}$. A duty cycle detect and divide circuit ($1/D$) 380 receives a high side control signal 310b and a low side control signal 320b. The signals 310b and 320b are derived from (related to) the signals 310 and 320, respectively, which control the switching on phase ($T_{sub.ON}$) and switching off phase ($T_{sub.OFF}$) but have a shorter dead time 322b than the dead time 322 for the signals 310 and 320 (see, FIG. 8) and are thus more accurately representative of the real duty cycle D of the switching regulator operation. The signal indicative of the sensed current $I_{sub.AVG}(LS)_{sub.sense}$ is then divided by the determined real duty cycle D to generate an output signal that is indicative of the extrapolated value $I_{sub.INextrap}$ for the input current $I_{sub.IN}$ of the switching regulator.

(84) Reference is now made to FIG. 10 which shows a circuit diagram for an embodiment of the current sensing circuit 340. A first input transistor M11 (for example, implemented as an n-channel MOS transistor) has a source terminal coupled, preferably directly connected, to the drain terminal

(d) of the LS transistor switch **314** (at the switching node **306**) and a drain terminal coupled, preferably directly connected, to a first (non-inverting (+)) input of a differential amplifier **342**. The gate terminal of the first input transistor **M11** is driven by the low side control signal **320**. A second input transistor **M12** (for example, implemented as an n-channel MOS transistor) has a source terminal coupled, preferably directly connected, to the source terminal (s) of the LS transistor switch **314** (at the reference node; ground) and a drain terminal coupled, preferably directly connected, to a second (inverting (-)) input of the differential amplifier **342**. The gate terminal of the second input transistor **M12** is also driven by the low side control signal **320**. Input transistors **M11** and **M12** are controlled by the low side control signal **320** to turn on during the switching on phase (T.sub.ON). An output transistor **M13** (for example, implemented as an n-channel MOS transistor) has a source terminal coupled, preferably directly connected, in a negative feedback circuit to the second (inverting) input of the differential amplifier **342**. A drain terminal of the output transistor **M13** is connected to an input leg of a current mirroring circuit **343** (formed by mirror connected p-channel MOS transistors). A resistor **344** has a first terminal coupled, preferably directly connected, to an output leg of the current mirroring circuit **343** and a second terminal coupled, preferably directly connected, to the reference voltage node (for example, ground). A passive low-pass filter circuit **348** formed by the series circuit connection of a resistor **350** and capacitor **352** is coupled, preferably directly connected, between the output leg of the current mirroring circuit **343** and the reference voltage node. A series connection node **356** of the low-pass filter circuit **348** between the resistor **350** and capacitor **352** provides an output of the current sensing circuit **340** producing the signal indicative of the sensed current

I.sub.AVG(LS).sub.sense. The first and second input transistors **M11**, **M12** function to sample the drain and source voltages, respectively, of the LS transistor switch **314** responsive to the low side control signal **320** only during switching on phase (T.sub.ON) for application to the differential amplifier **342**. The difference between those sampled voltages is converted by the differential amplifier **342** and output transistor **M13** to a current I.sub.sense which replicates the current flowing through the inductor **330** during switching on phase (T.sub.ON). This current I.sub.sense is mirrored by the current mirroring circuit **343** and converted to a voltage by resistor **344** and the voltage is smoothed by the low pass filter circuit **348** to generate an output voltage Vsense.

(85) FIG. 5B shows a circuit diagram for an embodiment of the differential amplifier **342**. A first amplifier transistor **M14** (for example, implemented as an n-channel MOS transistor) has a source terminal coupled, preferably directly connected, to the inverting (-) input of the differential amplifier **342** and a drain terminal biased by a first current source circuit **161** configured to source a current I.sub.bias from a supply voltage node. A second amplifier transistor **M15** (for example, implemented as an n-channel MOS transistor) has a source terminal coupled, preferably directly connected, to the non-inverting (+) input of the differential amplifier **342** and a drain terminal biased by a second current source circuit **163** configured to source the current I.sub.bias from the supply voltage node. The gate terminals of the transistors **M14** and **M15** are coupled, preferably directly connected, to each other and to the drain terminal of the transistor **M14**.

(86) Let K be a ratio of power to power sense. During the switching on phase (T.sub.ON) when the low side control signal **320** is asserted, transistors **314**, **M11** and **M12** are all turned on. If the gain of the differential amplifier **342** is high enough to ensure that the voltages at the inverting (-) input and non-inverting (+) input of the differential amplifier **342** are substantially equal (through the negative feedback with transistor **M13**), the following is true:

$$V_{\text{sub.306}} - K * R_{\text{sub.on}} * I_{\text{sub.bias}} = -K * R_{\text{sub.on}} * (I_{\text{sub.bias}} - I_{\text{sub.sense}})$$

(87) Where: V.sub.306 is the voltage at the switching node **306**, R.sub.on is drain-to-source resistance of the transistor **314** in the ON state (i.e., R_{ds_on}), I.sub.bias is the bias current for the current source circuits **161**, **163**, and I.sub.sense is the current flowing through the transistor **M13**.

(88) Then, the sense current I.sub.sense is:

$$(89) I_{\text{sense}} = \frac{V_{\text{sub.306}}}{K * R_{\text{on}}} = \frac{I_{\text{LS}}}{K}$$

(90) Where: $I_{sub,LS}$ is the current flowing through the inductor **330** when the transistor **314** in the ON state.

(91) It will accordingly be noted that the current $I_{sub,LS}$ and its replica current $I_{sub,sense}$ increase with a ratio of K . In this context, K is substantially constant, and it allows to have the current $I_{sub,sense}$ directly proportional to the current flowing through the inductor **330** when the LS transistor switch **314** is in the ON state. It is important that the value for K is large enough to have the magnitude of $I_{sub,sense}$ in a range of some micro-amperes, despite an inductor current $I_{sub,LS}$ that has a magnitude on the order of amperes.

(92) After mirroring by the current mirroring circuit **343**, the signal at the output of the current sensing circuit **340** that is indicative of the sensed current $I_{sub,AVG(LS).sub,sense}$ is actually a low pass filtered voltage V_{sense} that is equal to $R \cdot I_{sub,AVG(LS).sub,sense}$ where R is the resistance of the resistor **344**.

(93) Reference is now made to FIG. **11** which shows a circuit diagram for an embodiment of the duty cycle detect and divide circuit **380**. A differential amplifier **382** includes a non-inverting (+) input coupled to receive the voltage V_{sense} (which is indicative of the sensed current $I_{sub,AVG(LS).sub,sense}$ in the LS transistor switch **314** during the switching on phase ($T_{sub,ON}$)). The output of the differential amplifier **382** is coupled, preferably directly connected, to the inverting (−) input in negative feedback by a capacitor **384**. This forms a unity gain analog integration circuit that functions to integrate (i.e., average) the difference in voltages at the non-inverting (+) input and inverting (−) input. The voltage at the non-inverting (+) input is the voltage V_{sense} . The voltage at the inverting (−) input is a switched voltage selected by a voltage switching circuit **385** controlled by the ratio

(94) $\frac{T_{ON}}{T_{OFF}}$

to be equal to either the voltage $V_{sub,current_out}$ at the output of the differential amplifier **382** or the reference voltage (i.e., ground voltage).

(95) The voltage switching circuit **385** includes a first switching transistor **M16** (for example, implemented as a p-channel MOS transistor) having a source terminal coupled, preferably directly connected, to the output of the differential amplifier **382** and a drain terminal coupled, preferably directly connected, to a switching node **386**. The first switching transistor **M16** is controlled by a logical inverse of the signal **320b** to turn on during $T_{sub,ON}$ state (i.e., when transistor **314** is turned on by the low side control signal **320**). Thus, the gate of transistor **M16** may be configured to receive the logical inverse of the signal **320b** and is turned on when control signal **320b** is asserted (logic high). A second switching transistor **M17** (for example, implemented as an n-channel MOS transistor) has a drain terminal coupled, preferably directly connected, to the switching node **386** and a source terminal coupled, preferably directly connected, to the reference voltage node. The second switching transistor **M17** is controlled by a logical inverse of the signal **310b** during $T_{sub,OFF}$ state (i.e., when transistor **314** is turned off by the low side control signal **320**). Thus, the gate of transistor **M17** may be configured to receive the logical inverse of signal **310b** and is turned on when control signal **310b** is asserted (logic low). The switching node **386** is coupled in negative feedback through resistor **R1** to the inverting (−) input of the differential amplifier **382**. The signal generated at the switching node **386** is equal, on average, to $D \cdot V_{sub,current_out}$, where $V_{sub,current_out}$ is the voltage at the output of the differential amplifier **382**. When the first switching transistor **M16** is turned on during the switching on phase ($T_{sub,ON}$), the voltage $V_{sub,current_out}$ at the output of the differential amplifier **382** is applied to the inverting (−) input of the differential amplifier **382**. The differential amplifier **382** will respond to this by driving the voltage $V_{sub,current_out}$ toward the voltage $V_{sense} - V_{sub,current_out}$. Conversely, when the second switching transistor **M17** is turned on during the switching off phase ($T_{sub,OFF}$), the reference voltage (ground) is instead applied to the inverting (−) input of the differential amplifier **382**. The differential amplifier **382** will respond to this by driving the voltage $V_{sub,current_out}$ toward the voltage V_{sense} . Over time, due to the

averaging function provided by the unity gain integration on capacitor **384**, the voltage V.sub.current_out at the output of the differential amplifier **382** will settle at a value equal to:

$$(96) V_{\text{current_out}} = V_{\text{sense}} * \left(\frac{T_{\text{ON}} + T_{\text{OFF}}}{T_{\text{ON}}} \right) = \frac{V_{\text{sense}}}{D}$$

(97) In other words, supposing that the voltage at the (+) and (-) inputs of the amplifier **382** are equal, the following equation applies:

$$(98) V_{\text{current_out}} = V_{\text{sense}} * \frac{1}{D} = \frac{R}{K D} I_{\text{LS}} = \frac{R}{K} I_{\text{IN}}$$

(99) Where: R is the resistance of resistor **344** in FIG. **10**.

(100) The voltage V.sub.current_out at the output of the differential amplifier **382** is the signal indicative of the extrapolated value I.sub.INextrap for the input current I.sub.IN of the switching regulator.

(101) Reference is now made to FIG. **12** which shows a circuit diagram for another embodiment of the duty cycle detect and divide circuit **380**. A differential amplifier **392** includes a non-inverting (+) input coupled to receive the voltage V_{sense} (which is indicative of the sensed current I.sub.AVG(LS).sub.sense in the LS transistor switch **314** during the switching on phase (T.sub.ON)). The output of the differential amplifier **392** is coupled, preferably directly connected, to the inverting (-) input in negative feedback by a resistor **394** having a fixed resistance R_b. The inverting (-) input is further coupled to ground through a resistor **396** having a variable resistance R_a. The value of the variable resistance R_a is digitally controlled by the bits of a digital selection signal **398** generated by an analog-to-digital converter (ADC) circuit **400**. This circuit is essentially a non-inverting operational amplifier with a controllable gain dependent on the resistances R_a and R_b (where the amplifier gain

$$(102) G = \left(1 + \frac{R_b}{R_a} \right).$$

(103) A voltage switching circuit **402** includes a first switching transistor M**18** (for example, implemented as a p-channel MOS transistor) having a source terminal coupled, preferably directly connected, to a supply voltage V_{dd} and a drain terminal coupled, preferably directly connected, to a switching node **304**. The first switching transistor M**18** is controlled by a logical inverse of signal **320b** to turn on during T.sub.ON state (i.e., when transistor **314** is turned on by the low side control signal **320**). Thus, the gate of transistor M**18** may be configured to receive the logical inverse of signal **320b** and is turned on when control signal **320b** is asserted (logic high). A second switching transistor M**19** (for example, implemented as an n-channel MOS transistor) has a drain terminal coupled, preferably directly connected, to the switching node **304** and a source terminal coupled, preferably directly connected, to the reference voltage node (ground). The second switching transistor M**19** is controlled by a logical inverse of signal **310b** to turn on during T.sub.OFF state (i.e., when transistor **314** is turned off by the low side control signal **320**). Thus, the gate of transistor M**19** may be configured to receive the logical inverse of signal **310b** and is turned on when control signal **310b** is asserted (logic low). The switching node **304** is coupled to the input of the ADC circuit **400** through a low-pass filter circuit formed by resistor **306** and capacitor **308**.

(104) The voltage at the switching node **304** is a switched voltage selected by the voltage switching circuit **402** controlled by the ratio

$$(105) \frac{T_{\text{ON}}}{T_{\text{OFF}}}$$

to be equal to either the supply voltage V_{dd} or the reference voltage (i.e., ground voltage). Analog filtering of the switched voltage by the low-pass filter circuit produces an input voltage to the ADC circuit **400** equal, on average, to D*V_{dd}. This switched and filtered voltage is converted by the ADC circuit **400** to generate the digital selection signal **398** for controlling the variable resistance R_a of the resistor **396**. The resistances R_a, R_b of the resistors **396**, **394** set the gain of the differential amplifier **392** applied to the voltage V_{sense} (indicative of the sensed current I.sub.AVG(LS).sub.sense in the LS transistor switch **314** during the switching on phase (T.sub.ON)) which is applied to the non-inverting (+) input.

(106) The following relationship is achieved through the non-inverting amplifier circuit using the

differential amplifier **392** with resistors **394** and **396**:

$$(107) V_{\text{current_out}} = V_{\text{sense}}(1 + \frac{R_b}{R_a})$$

(108) The variable resistance R_a of the resistor **396** is digitally controlled to satisfy the following relationship:

$$(109) 1 + \frac{R_b}{R_a} = \frac{1}{D}$$

(110) Because R_a must satisfy the foregoing equation, this means that:

$$(111) R_a = \frac{D}{1-D} R_b$$

(112) Where the resistance R_b is fixed. If we consider, as an example, the ADC circuit **400** with a three-bit resolution, the input signal $D \cdot V_{\text{dd}}$ is then quantized to eight levels, with the following Table defining the relationship of the variable resistance R_a as a function of the duty cycle and the fixed resistance R_b :

(113) TABLE-US-00002

D	R_a
0	0
1/8	$R_b/7$
2/8	$R_b/3$
3/8	$3 \cdot R_b/5$
4/8	R_b
5/8	$5 \cdot R_b/3$
6/8	$3 \cdot R_b$
7/8	$7 \cdot R_b$

(114) The output of the ADC circuit **400** controls the variable resistor **396** to select the correct resistance R_a . This can be accomplished, for example, using an analog multiplexer having inputs connected to resistances for the possible values of R_a , a control input connected to the output of the ADC circuit **400** and an output connected to resistor R_b and the input of amplifier **392**.

(115) It will be noted that the implementation with a three-bit resolution is just an example, and this concept is extendible to a higher resolution to provide higher accuracy in the selection of the variable resistance value.

(116) Because of the foregoing relationship between D , R_a and R_b , the following equation applies:

$$(117) 0V_{\text{current_out}} = V_{\text{sense}} * \frac{1}{D} = \frac{R}{K D} I_{\text{LS}} = \frac{R}{K} I_{\text{IN}}$$

(118) Where: R is the resistance of resistor **344** of FIG. **10**.

(119) The voltage $V_{\text{sub.current_out}}$ at the output of the differential amplifier **392** is the signal indicative of the extrapolated value $I_{\text{sub.IN}}^{\text{extrap}}$ for the input current $I_{\text{sub.IN}}$ of the switching regulator.

(120) Reference is once again made to FIG. **8**. The output current $I_{\text{sub.OUT}}$ of the switching regulator is the current that flow through the high side transistor **304** to the load. This current is the difference between the input current $I_{\text{sub.IN}}$ of the switching regulator and the current $I_{\text{sub.LS}}$ of the low side transistor **314**. To extract a value for the output current $I_{\text{sub.OUT}}$, and generate a signal indicative of the extrapolated value $I_{\text{sub.OUT}}^{\text{extrap}}$ for the output current $I_{\text{sub.OUT}}$ of the switching regulator, a summation circuit subtracts the sensed current $I_{\text{sub.AVG(LS).sub.sense}}$ in the LS transistor switch **314** from the extrapolated value $I_{\text{sub.IN}}^{\text{extrap}}$ for the input current $I_{\text{sub.IN}}$.

(121) FIG. **9** illustrates operation of the boost converter in continuous conduction mode (CCM). It will, however, be noted that the current sensing circuit **340** and duty cycle detect and divide circuit **380** can also operate for sensing inductor input current during ON time where the converter is instead configured for operation in discontinuous conduction mode (DCM). In this case, beyond the ON and OFF times, a high-impedance state is present when the coil current is zero. In this high-impedance state: transistors **304**, **314** are turned off, transistors **M16**, **M17** are turned off, and transistors **M18**, **M19** are turned off.

(122) Although illustrated herein by example only for the calculation of switching regulator output current for a buck-type circuit (FIG. **2**) and the calculation of switching regulator input current for a boost-type circuit (FIG. **8**), it will be understood that the technique disclosure herein is equally applicable to current detection in other types of switching regulator circuits (buck-boost, inverting, etc.).

(123) While the invention has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are considered illustrative or exemplary and not restrictive; the invention is not limited to the disclosed embodiments. Other variations to the

disclosed embodiments can be understood and effected by those skilled in the art in practicing the claimed invention, from a study of the drawings, the disclosure, and the appended claims.

Claims

1. A switching regulator circuit, comprising: a high side (HS) transistor coupled between a first node and a switching node; a low side (LS) transistor coupled between the switching node and a second node; driver circuitry for controlling actuation of the HS transistor and LS transistor in accordance with a duty cycle including a switching on phase and a switching off phase with a first dead time delimiting the switching off phase; an inductor having a first terminal coupled to the switching node and a second terminal coupled to a third node, wherein a coil current flows through the inductor; and a coil current detection circuit comprising: a current sensing circuit configured to sense a transistor current flowing through only one of the HS transistor and LS transistor during the switching on phase and generate a first signal indicative of the sensed transistor current; and a duty cycle detect and divide circuit configured to detect said duty cycle which controls actuation of the HS transistor and LS transistor, and divide the first signal indicative of the sensed transistor current by said duty cycle to generate an output signal indicative of the coil current that flows through the inductor.
2. The circuit of claim 1, wherein the switching regulator circuit is a buck-type regulator circuit, wherein the first node is configured to receive an input voltage, the second node is configured to receive a reference voltage and the third node is an output voltage node, and wherein the coil current is output current from the buck-type regulator circuit.
3. The circuit of claim 1, wherein the switching regulator circuit is a boost-type regulator circuit, wherein the first node is an output voltage node, the second node is configured to receive a reference voltage and the third node is configured to receive an input voltage, and wherein the coil current is input current to the boost-type regulator circuit.
4. The circuit of claim 1, wherein said duty cycle detect and divide circuit comprises: a first differential amplifier having a first input coupled to receive the first signal indicative of the sensed transistor current and a second input and an output producing said output signal; a capacitor coupled in feedback between the output of the first differential amplifier and the second input; a first switching transistor coupled between the output of the first differential amplifier and a further switching node, said first switching transistor controlled to turn on corresponding to the switching on phase for the duty cycle; a second switching transistor coupled between the further switching node and the second node, said second switching transistor controlled to turn on corresponding to the switching off phase for the duty cycle; and a feedback path coupled between the further switching node and the second input of the first differential amplifier.
5. The circuit of claim 4, wherein the first and second switching transistors are controlled by signals have a second dead time shorter than the first dead time.
6. The circuit of claim 1, wherein said duty cycle detect and divide circuit comprises: a first differential amplifier having a first input coupled to receive the first signal indicative of the sensed transistor current and a second input and an output producing said output signal; a capacitor coupled in feedback between the output of the first differential amplifier and the second input; and a switching feedback circuit controlled in response to alternation between the switching on phase and switching off phase with a second dead time shorter than the first dead time to alternately apply a signal at the output of the first differential amplifier and a reference voltage to the second input of the first differential amplifier.
7. The circuit of claim 1, wherein said duty cycle detect and divide circuit comprises: a first differential amplifier having a first input coupled to receive the first signal indicative of the sensed transistor current and a second input and an output producing said output signal; a first resistor having a fixed resistance R_b coupled in feedback between the output of the first differential

amplifier and the second input; a second resistor having a variable resistance R_a coupled between the second input and the second node; a first switching transistor coupled between a supply voltage node and a further switching node, said first switching transistor controlled to turn on corresponding to the switching on phase for the duty cycle; a second switching transistor coupled between the further switching node and the second node, said second switching transistor controlled to turn on corresponding to the switching off phase for the duty cycle; and an analog-to-digital converter circuit configured to convert a voltage output from the further switching node to a digital control signal for controlling selection of the variable resistance R_a of the second resistor.

8. The circuit of claim 7, wherein the digital control signal selects the variable resistance R_a to satisfy the following equation: $1 + \frac{R_b}{R_a} = \frac{1}{D}$ where D is the duty cycle.

9. The circuit of claim 7, wherein the first and second switching transistors are controlled by signals have a second dead time shorter than the first dead time.

10. The circuit of claim 1, wherein said current sensing circuit comprises: a second differential amplifier having a first input, a second input and an output; a first input transistor coupled between a first terminal of said one of the HS transistor and LS transistor and the first input of the second differential amplifier; a second input transistor coupled between a second terminal of said one of the HS transistor and LS transistor and the second input of the second differential amplifier; wherein said first and second input transistors are actuated during switching on phase for the duty cycle; an output transistor coupled between the first input of the second differential amplifier and a first intermediate node; and a resistor coupled between the first intermediate node and the second node; wherein said first signal indicative of the sensed transistor current is derived from a voltage across said resistor.

11. The circuit of claim 10, wherein said current sensing circuit further comprises a low pass filter circuit coupled between the first intermediate node and the second node, said low pass filter circuit having a second intermediate node outputting the first signal indicative of the sensed transistor current.

12. The circuit of claim 1, wherein said current sensing circuit comprises: a second differential amplifier having a first input, a second input and an output; a first input transistor coupled between a first terminal of said one of the HS transistor and LS transistor and the first input of the second differential amplifier; a second input transistor coupled between a second terminal of said one of the HS transistor and LS transistor and the second input of the second differential amplifier; wherein said first and second input transistors are actuated during switching on phase for the duty cycle; an output transistor coupled between the first input of the second differential amplifier and a first intermediate node; a current mirror having an input leg coupled to the first intermediate node and an output leg coupled to a second intermediate node; and a resistor coupled between the second intermediate node and the second node; wherein said first signal indicative of the sensed transistor current is derived from a voltage across said resistor.

13. The circuit of claim 12, wherein said current sensing circuit further comprises a low pass filter circuit coupled between the second intermediate node and the second node, said low pass filter circuit having a third intermediate node outputting the first signal indicative of the sensed transistor current.

14. A circuit for detecting a coil current flowing through an inductor of a switching regulator circuit having a switching transistor coupled to the inductor and driver circuitry for controlling actuation of the switching transistor in accordance with a duty cycle including a switching on phase and a switching off phase delimited by dead time, comprising: a current sensing circuit configured to sense a transistor current flowing through the switching transistor during switching on phase of the duty cycle to generate a first signal indicative of the sensed transistor current; and a duty cycle detect and divide circuit configured to detect said duty cycle, and divide the first signal indicative of the sensed transistor current by said duty cycle to generate an output signal indicative of the coil current flowing through the inductor of the switching regulator circuit.

15. The circuit of claim 14, wherein the switching regulator circuit is a buck-type regulator circuit, wherein the switching transistor is a high-side transistor, and wherein the coil current is output current from the buck-type regulator circuit.
16. The circuit of claim 14, wherein the switching regulator circuit is a boost-type regulator circuit, wherein the switching transistor is a low-side transistor, and wherein the coil current is input current to the boost-type regulator circuit.
17. The circuit of claim 14, wherein said duty cycle detect and divide circuit comprises: a first differential amplifier having a first input coupled to receive the first signal indicative of the sensed transistor current and a second input and an output producing said output signal; a capacitor coupled in feedback between the output of the first differential amplifier and the second input; and a switching feedback circuit controlled in response to the switching on phase and switching off phase of said duty cycle to alternately apply a signal at the output of the first differential amplifier and a reference voltage to the second input of the first differential amplifier.
18. The circuit of claim 14, wherein said duty cycle detect and divide circuit comprises: a first differential amplifier having a first input coupled to receive the first signal indicative of the sensed transistor current and a second input and an output producing said output signal; a first resistor having a fixed resistance R_b coupled in feedback between the output of the first differential amplifier and the second input; a second resistor having a variable resistance R_a coupled between the second input and the second node; an analog-to-digital converter circuit configured to convert a control voltage to a digital control signal for controlling selection of the variable resistance R_a of the second resistor; and a switching feedback circuit controlled in response to the switching on phase and switching off phase of said duty cycle to alternately select a supply voltage and a ground voltage in generating said control voltage.
19. The circuit of claim 18, wherein the digital control signal selects the variable resistance R_a to satisfy the following equation: $1 + \frac{R_b}{R_a} = \frac{1}{D}$ where D is the duty cycle.
20. The circuit of claim 14, wherein said current sensing circuit comprises: a second differential amplifier having a first input, a second input and an output; a first input transistor coupled between a first terminal of said switching transistor and the first input of the second differential amplifier; a second input transistor coupled between a second terminal of said switching transistor and the second input of the second differential amplifier; wherein said first and second input transistors are actuated during switching on phase for the duty cycle; and an output transistor coupled between the first input of the second differential amplifier and a first intermediate node to generate said first signal indicative of the sensed transistor current.
21. A method for detecting a coil current flowing through an inductor of a switching regulator circuit having a switching transistor coupled to the inductor and driver circuitry for controlling actuation of the switching transistor in accordance with a duty cycle including a switching on phase and a switching off phase delimited by dead time, comprising: sensing a transistor current flowing through the switching transistor during switching on phase to generate a first signal indicative of the sensed transistor current; detecting said duty cycle; and dividing the first signal indicative of the sensed transistor current by said duty cycle to generate an output signal indicative of the coil current flowing through the inductor of the switching regulator circuit.
22. The method of claim 21, wherein the switching regulator circuit is a buck-type regulator circuit, wherein the switching transistor is a high-side transistor, and wherein the coil current is output current from the buck-type regulator circuit.
23. The method of claim 21, wherein the switching regulator circuit is a boost-type regulator circuit, wherein the switching transistor is a low-side transistor, and wherein the coil current is input current to the boost-type regulator circuit.
24. The method of claim 21, wherein sensing the transistor current comprises: sampling a first voltage at a first terminal of the switching transistor during switching on phase; sampling a second voltage at a second terminal of the switching transistor during switching on phase; differentially

processing the first and second voltages to generate a sensed current replicating the transistor current flowing through the switching transistor; converting the sensed current to a voltage; and smoothing the voltage to generate said first signal indicative of the sensed transistor current.

25. The method of claim 21, wherein detecting and dividing comprises: integrating a difference between the first signal indicative of the sensed transistor current and a switched signal; wherein said switched signal has a first value during the switching on phase of the duty cycle and wherein the switched signal has a second value during the switching off phase of the duty cycle; wherein the first value is the output signal and the second value is a reference value.

26. The method of claim 21, wherein detecting and dividing comprises: amplifying the first signal indicative of the sensed transistor current by a gain value generate the output signal; generating a switched signal having a first value during the switching on phase of the duty cycle and a second value during the switching off phase of the duty cycle; filtering and converting the switched signal to a digital control signal; selecting the gain value using the digital control signal.

27. The method of claim 26, wherein the first value is a supply voltage value and the second value is a reference value.
